

One-to-one transformations

Definition (injective linear transformation)

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation, and let $\vec{x}_1$ and $\vec{x}_2$ be in $\mathbb{R}^n$. We say that T is an **injection** or is **one-to-one** if $\vec{x}_1 \neq \vec{x}_2$ implies that $T(\vec{x}_1) \neq T(\vec{x}_2)$. Equivalently, if $T(\vec{x}_1) = T(\vec{x}_2)$, then $\vec{x}_1 = \vec{x}_2$.

Theorem

Let A be an $m \times n$ matrix and let $\vec{x}$ be a vector of length n . Then the transformation induced by A , T_A , is one-to-one if and only if $A\vec{x} = \vec{0}$ implies $\vec{x} = \vec{0}$.

(2)

(1)

(1) $\Rightarrow$ (2): Let $\vec{u}$ be an element of $\mathbb{R}^n$ such that $T_A(\vec{u}) = \vec{0}$, then since T_A is a **one-to-one linear transformation**, we have $T_A(\vec{0}) = \vec{0}$ and $\vec{0} = \vec{u}$.

(2) $\Rightarrow$ (1): Let $\vec{u}$ and $\vec{v}$ be elements in $\mathbb{R}^n$ such that $A\vec{u} = A\vec{v}$. Then $A(\vec{u} - \vec{v}) = \vec{0}$ which implies $\vec{u} - \vec{v} = \vec{0}$ or $\vec{u} = \vec{v}$. Hence, we have shown that if $T_A(\vec{u}) = T_A(\vec{v})$ then $\vec{u} = \vec{v}$, which means T_A is one-to-one.

Question: True/False?

Let $T(\vec{x}) = A\vec{x}$, where A is an $m \times n$ matrix. If we also assume that T is injective, then which statement is true and which one is false?

(I) For every $\vec{b} \in \mathbb{R}^m$, there exists one solution for the equation $T(\vec{x}) = \vec{b}$.

False. For example, let $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$ and $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$, which is a one-to-one transformation. Now, for $\vec{b} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

the system $A\vec{x} = \vec{b}$ has no solution, since the augmented matrix is

$$\left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & 0 & 3 \end{array} \right] \xrightarrow{R_3 - R_1} \left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{array} \right] \text{ and therefore the SLE is inconsistent.}$$

(II) $A\vec{x} = \vec{0}$ has only the trivial solution.

This is True. See the theorem that was proved last session.

(III) The columns of A are linearly independent. True.

$$A\vec{x} = \vec{0} \iff x_1\vec{A}_1 + x_2\vec{A}_2 + \dots + x_n\vec{A}_n = \vec{0}$$

If $A\vec{x} = \vec{0}$ has only the trivial solution, it means that none of

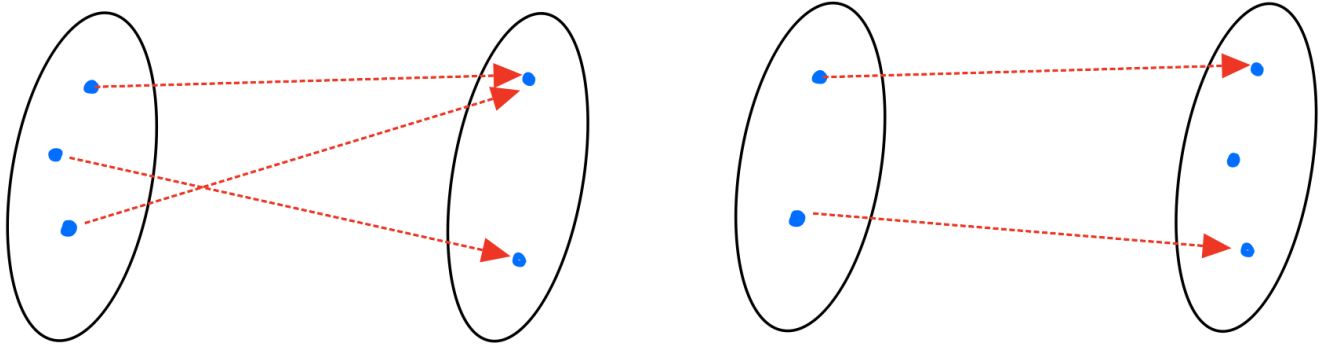
the columns of A can be written as a linear combination of the rest of them

since if for example $\vec{A}_1$ can be written as a linear combination:

$$\vec{A}_1 = a_2\vec{A}_2 + a_3\vec{A}_3 + \dots + a_n\vec{A}_n, \text{ which gives } \vec{A}_1 - a_2\vec{A}_2 - a_3\vec{A}_3 - \dots - a_n\vec{A}_n = \vec{0}.$$

This contradicts the fact that $A\vec{x} = \vec{0}$ has only the trivial solution.

Which transformation is an injection (one-to-one)?



Which transformation is an injection (one-to-one)?

Ⓐ Reflection in the x -axis in $\mathbb{R}^2$.

The corresponding matrix is $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ and the SLE $A\vec{x} = \vec{0}$ has only the trivial solution, hence this is an injection.

Ⓑ Projection into the x -axis in $\mathbb{R}^2$.

Similarly $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, whereas $A\vec{x} = \vec{0}$ has non-trivial solutions, hence this is NOT an injection.

Ⓒ $T \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$

Again $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$, however, $A\vec{x} = \vec{0}$ has a free parameter and

Therefore Ⓒ is not an injection. For example, $T \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = T \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

Surjections

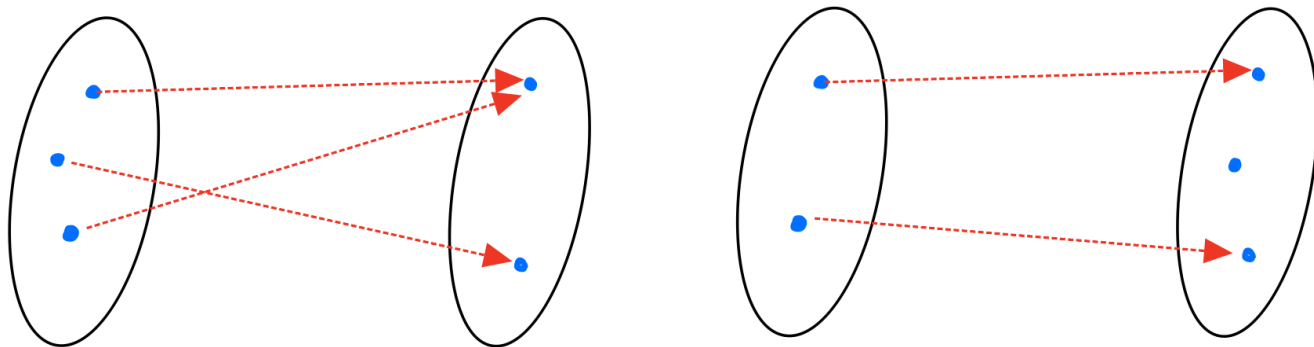
Definition

Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. We say that T is a **surjection** or **onto** if, for every $\vec{b} \in \mathbb{R}^m$ there exists an $\vec{x} \in \mathbb{R}^n$ such that $T(\vec{x}) = \vec{b}$.

Problem

Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation defined by $T \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} a+b \\ 0 \end{bmatrix}$ for all $\begin{bmatrix} a \\ b \end{bmatrix} \in \mathbb{R}^2$. Determine whether T is a surjective transformation or not.

Which transformation is a surjection (onto)?



Which transformation is a surjection (onto)?

$$(I) \quad T: \mathbb{R}^3 \rightarrow \mathbb{R}^2, \quad T(\vec{x}) = \begin{bmatrix} x-z \\ y+z \end{bmatrix} \quad A = [T(\vec{e}_1) \quad T(\vec{e}_2) \quad T(\vec{e}_3)] = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$A\vec{x} = \vec{b} \quad \text{or} \quad \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \quad \text{will always have a solution}$$

Since $\left[\begin{array}{ccc|c} 1 & 0 & -1 & b_1 \\ 0 & 1 & 1 & b_2 \end{array} \right]$ has two leading ones and 3 variables, one of which is a free variable. Hence, T is a surjection.

$$(II) \quad T(\vec{x}) = A\vec{x}, \text{ where } A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

$$A\vec{x} = \vec{b} \text{ or } \left[\begin{array}{cc|c} 1 & 0 & b_1 \\ 0 & 1 & b_2 \\ 0 & 0 & b_3 \end{array} \right] \text{ has no solution for } b_3 \neq 0, \text{ which}$$

means T is not a Surjection.